# A3-PS016-TC106

## A Simulation-Based Handover Decision Evaluation for High-Mobility Users

**Nabhonil Bhattacharjee**

**Arka Manna**

nabhonilbhattacharjee@gmail.com | arkamanna20@gmail.com

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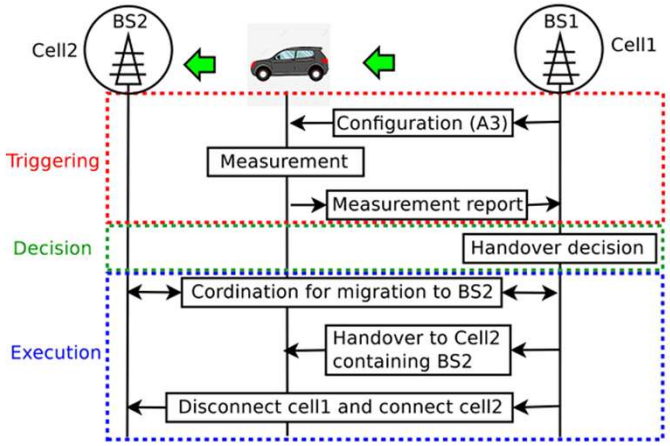
# Problem Description

- The core challenge is evaluating **learned handover decision policies** against **conventional signal-based baselines** for high-mobility users across multiple cells.
- High-speed mobile user equipment (UE) traversing dense **5G heterogeneous networks** (HetNets) **suffers from frequent and unnecessary handovers**.
- Current handover mechanisms **rely heavily on static metrics**, which cause high rates of handover ping-pong (HOPP), radio link failures (RLF), and handover failures (HOF) during rapid movement.
- Poorly timed handovers in dynamic scenarios **degrade quality of service** (QoS), leading to connection drops, increased latency, and service interruption.
- Testing handover strategies at scale is difficult, making reproducible cellular-network simulators essential for bridging the gap between algorithm development and real-world validation.
- **Future Applications & Importance**: Optimizing these decisions is critical for maintaining seamless connectivity in the **future of autonomous vehicles, high-speed rail communications, High Fidelity Drone Controls and low-latency augmented reality** in 5G/6G networks.

# Background

- The advent of 5G and B5G introduces dense **HetNets** comprising **macrocells, microcells, picocells, and femtocells** to meet high data demands.
- These deployments mix **sub-6 GHz (FR1)** and **mmWave (FR2)** bands, drastically changing signal propagation and increasing sensitivity to physical blockages.
- Traditional handover execution relies primarily on **Reference Signal Received Power (RSRP)** thresholds alongside static **Time-to-Trigger (TTT)** and **Hysteresis Margin (HOM)** parameters.
- As UE velocity increases, standard signal-based handovers trigger too slowly or too frequently, failing to adapt to rapidly changing channel conditions.
- Simulation-based evaluation is required to safely test machine-learning-driven handover control policies across multi-seed environments and varying network loads before real-world deployment.

| Event | Criteria | Explanation |
|---|---|---|
| A1 | $R_s > \Delta_{A1}$ | Serving cell's RSRP is better than a threshold |
| A2 | $R_s < \Delta_{A2}$ | Serving cell's RSRP is worse than a threshold |
| A3 (A6) | $R_n > R_s + \Delta_{A3}$ | Neighbor cell is better than serving cell with a offset |
| A4 (B1) | $R_n > \Delta_{A4}$ | Neighbor cell is better than a threshold |
| A5 (B2) | $R_s < \Delta^1_{A5'}$ $R_n > \Delta^2_{A5}$ | Serving cell is worse than a threshold value, and neighbor cell is better than a threshold |



# Literature & RFC Review

- **Standard Protocols & RFCs:** RFC 2330 defines the framework for IP Performance Metrics, establishing standardized measurement criteria for network evaluation.
- **3GPP Handover Standards:** Existing 3GPP Rel.15 (A3 event-triggered) and Rel.16 Conditional Handover (CHO) rely on static RSRP metrics and decouple preparation from execution, but lack real-time environmental adaptability.
- **Machine Learning Integration:** Recent proposals utilize Reinforcement Learning (RL) algorithms like SARSA to dynamically adapt TTT and Hysteresis based on predicted signal quality.
- **Proactive Decision Making (PDM):** PDM approaches use machine learning to predict and select target cells based on network download rates prior to the handover trigger, reducing delay.
- **Cell Identity Filtering:** Solutions using NR Cell Identity (NCI) coding help high-speed UEs avoid highly unstable mmWave cells in favor of more reliable sub-6GHz macro/small cells.
- **Limitations of Existing Studies:** Many current research models rely on mathematical abstractions or simplified simulators that do not adequately reflect real-world physical channel constraints or struggle with scalability in multi-cell high-mobility environments.

# Proposed Solution

- We will develop an end-to-end 5G HetNet Simulator utilizing ns-3 and the 5G-LENA module to create a strictly reproducible cellular-simulation environment.
- Our architecture will successfully decouple the physical-layer simulation from the algorithmic evaluation layer.
- We introduce a learned handover decision model implemented via a Python-based handover simulator that consumes generated PHY/MAC traces to evaluate our candidate decisions.
- Our supervised learning model evaluates RSRP, bandwidth, and mobility patterns to greedily select the optimal cell and optimize Handover Control Parameters dynamically.
- The solution evaluates 3 cells and 2 user-mobility speeds over multiple random seeds, tracking key performance indicators such as handover failures, ping-pong events, interruption duration, and throughput.
- Our custom framework compares this learned policy directly against a conventional signal-based baseline under identical conditions to prove performance gains.

# Optimization Proposed

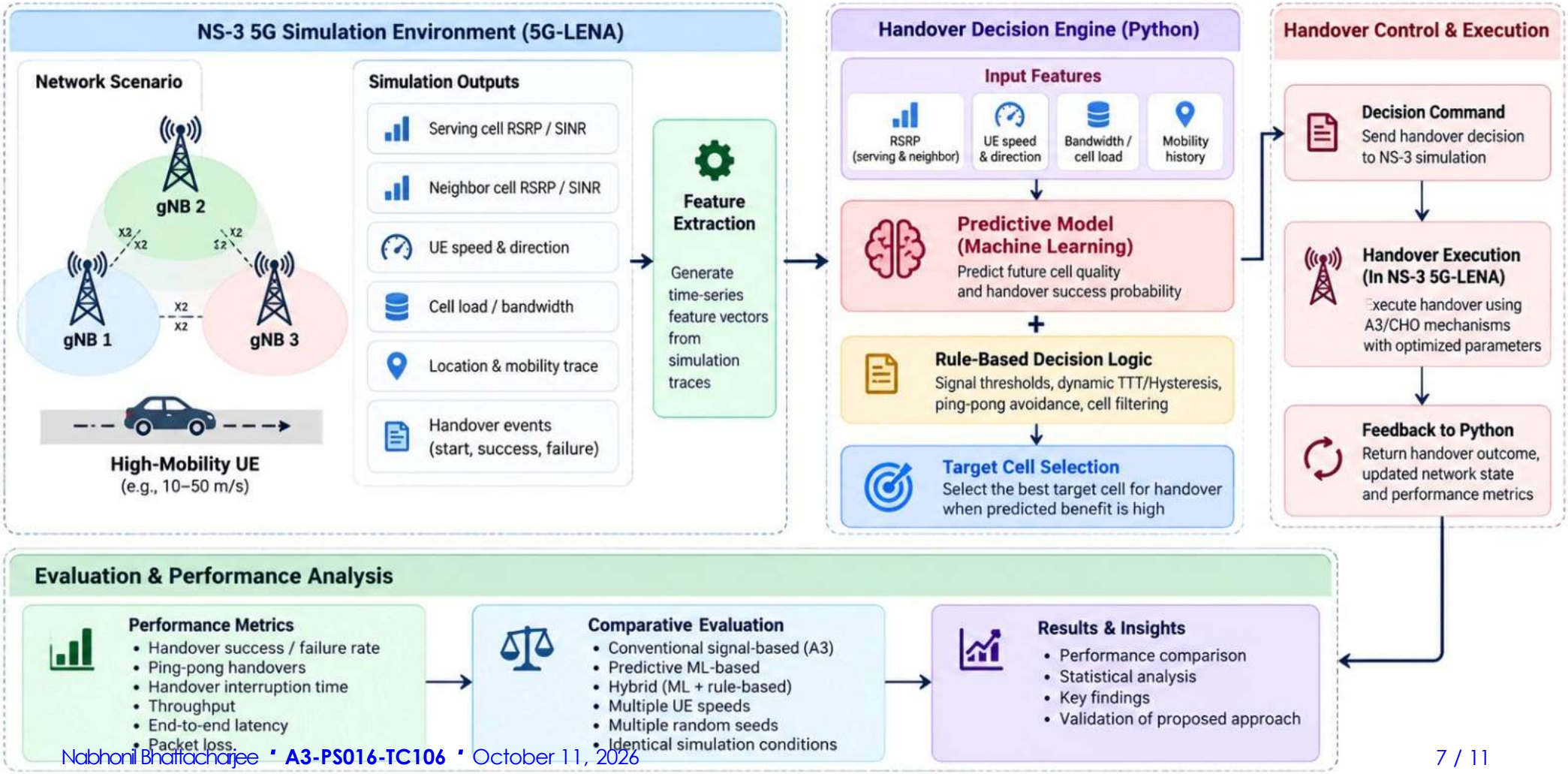
- **Dynamic Parameter Tuning:** Real-time adaptation of Time-to-Trigger (TTT) and Hysteresis parameters based on user velocity and predicted signal degradation.
- **Predictive Cell Filtering:** Filtering out unstable mmWave candidate cells for high-speed users to ensure connection continuity.
- **Score-Based Evaluation:** Ranking candidate cells using a combined score of RSRP and available bandwidth rather than solely relying on signal strength.

| Feature | Before (Traditional Baseline) | After (Proposed Optimization) |
|---|---|---|
| Decision Metric | Static RSRP thresholds | Learned policy scoring RSRP, speed, and cell load |
| Handover Parameters | Fixed TTT and Hysteresis | Dynamically adapted TTT and Hysteresis |
| High-Mobility Handling | Prone to ping-ponging and failures | Predictive cell filtering to maintain stability |
| Evaluation Method | Costly real-world trials or simplified mathematical models | Reproducible full-stack cellular simulation |

# 7 Solution Architecture

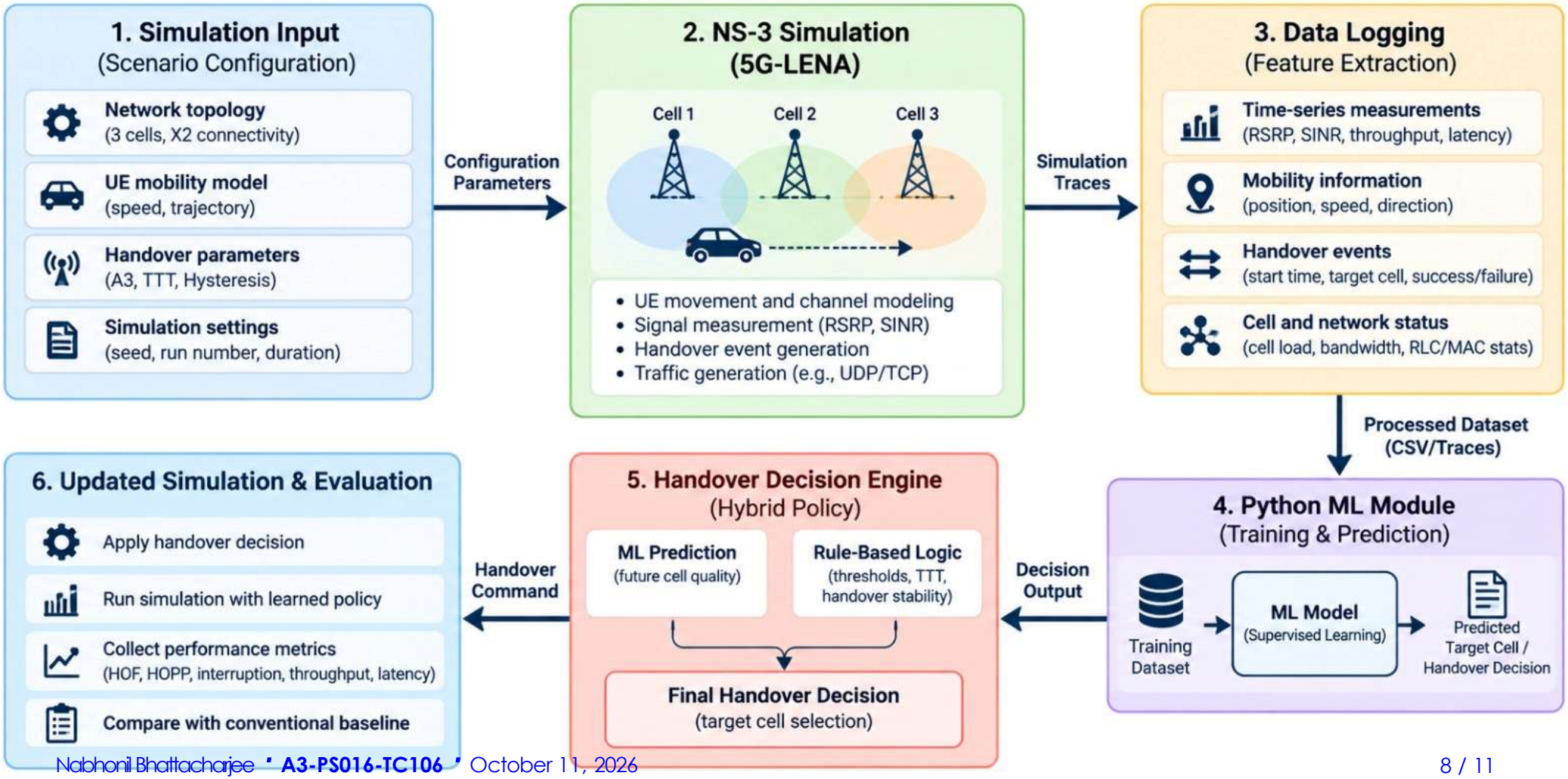
AIORI 3 Architecture Design (UML Diagram)

Predictive Handover Decision Framework for High-Mobility Users



# 8 Data Flow Diagram

PS-016: High-Mobility Handover Decision



# Deployment Timeline

**Phase 1: Foundation & Setup:** Configure GitHub collaboration, establish the ns-3 / 5G-LENA environment, and define 3 cells with 2 user-mobility speeds. (16th to 19th October)
**Phase 2: Baseline Implementation:** Implement the conventional signal-based handover baseline and generate the initial PHY/MAC traces. (20th to 23rd October)
**Phase 3: Model Integration & Testing:** Deploy the learned handover decision model, executing runs across multiple random seeds to measure failures, ping-pongs, throughput, and interruptions. (24th to 27th October)
**Phase 4: Final Submission:** Finalize performance analysis visualizations, upload the technical report, and present the prototype demonstration. (28th to 31st October)

# Bibliography I


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2. A. Priyanka et al., "Machine learning algorithms in proactive decision making for handover management from 5G & beyond 5G," *Egyptian Informatics Journal*, vol. 24, p. 100389, 2023.
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# Thank You for Your Attention!

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